Thermodynamic description of the Pb–Zn system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

Synthetic document generated by the ORCHID pipeline benchmark; not a real publication. CC BY 4.0. Derived from the SGTE binary dataset collection (B. Hallstedt, Calphad 89 (2025) 102833, CC BY 4.0) and the original assessment cited in the dataset.

Abstract

This synthetic document describes the Pb–Zn binary system as calculated with pycalphad from the SGTE dataset PbZn-98Jan.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L1 + L2 → (Zn), lies at 420.0 °C.

1. Introduction

The Pb–Zn system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as miscibility gap, monotectic. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of T. Jantzen, P.J. Spencer, Calphad, 22, 417-34(1998). [2]. The calculations use pycalphad 0.11.2 at 1 atm.

For Pb, the liquidus meets the axis at 327.5 °C. The melting point of Zn reproduced by the dataset is 419.5 °C.

2. Phase diagram and invariant reactions

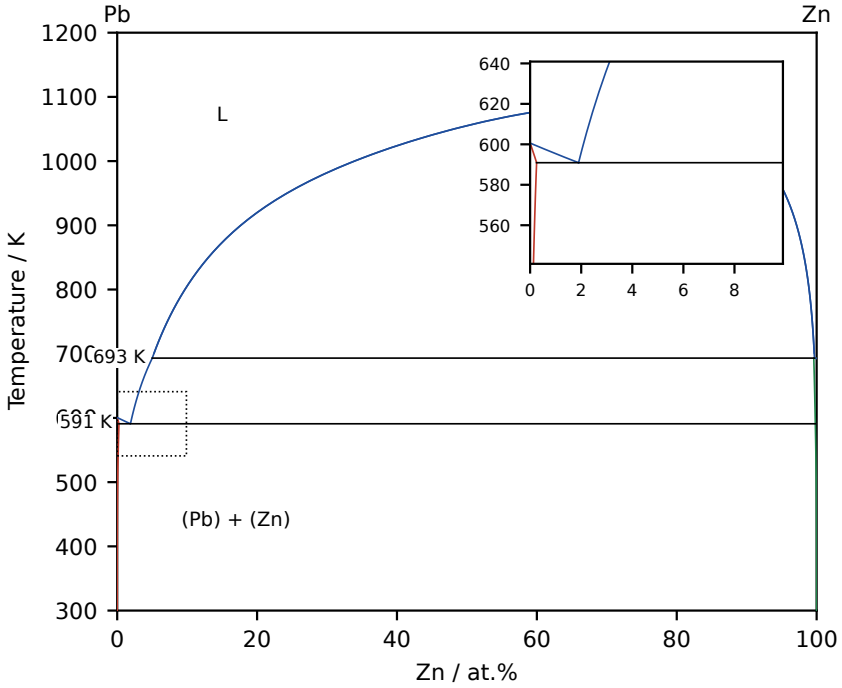


Fig. 1. Calculated Pb–Zn phase diagram (inset: enlarged view near the lowest invariant).

We obtain 420.0 °C for the syntectic reaction (L1 + L2 → (Zn)), where (Zn) has 99.6 at.% Zn. Eutectic equilibrium L → (Pb) + (Zn) is found at 317.8 °C and 1.9 at.% (L).

Table 1 compares the calculated temperatures with experimental values from the literature [3–5]; in this synthetic document those values and references are fictitious.

Table 1. Invariant reactions and congruent points, calculated and experimental (Pb–Zn).

Reaction	Type	T (°C) this work	T exp.[b]	at.% Zn, 1	at.% Zn, 2	at.% Zn, 3
L1 + L2 → (Zn)	syntectic	420.0	429.0 [3]	5.0	99.7	99.6
L → (Pb) + (Zn)	eutectic	317.8	321.8 [5]	1.9	0.3	99.8

[b] Experimental temperatures are fictitious values attributed to invented references [3–5] for testing evidence-class extraction; they are not data.

3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Pb–Zn).

	Structure			Stability range / °C		Homogeneity range	
Phase	Pearson	Strukturb.	Prototype	from	to	x(Zn)	at T / °C
(Pb)	cF4	A1	Cu	< 27[a]	327.5	0.000–0.003	317
(Zn)	hP2	A3	Mg	< 27[a]	420.0	0.996–1.000	417

[a] Stable at the lowest calculated temperature.

The upper stability limit of (Zn) is 420.0 °C.

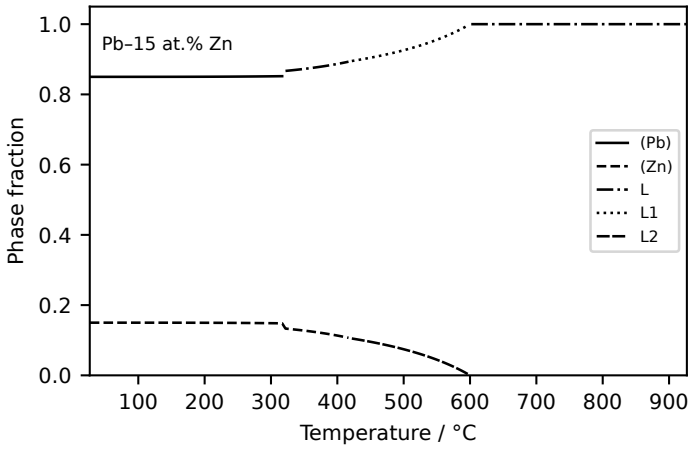


Fig. 2. Calculated equilibrium phase fractions versus temperature for Pb–15 at.% Zn.

4. Thermodynamic properties

At 1150 K the integral mixing enthalpy of the liquid has its maximum, 8.49 kJ/mol, at 60.0 at.% Zn. For the equiatomic liquid at 1150 K, the Gibbs energy of mixing is −2.33 kJ/mol-atom.

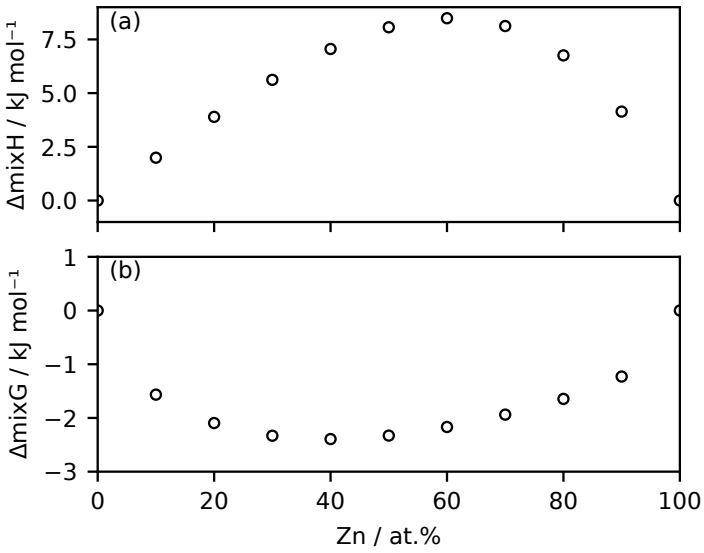


Fig. 3. Calculated (a) mixing enthalpy and (b) mixing Gibbs energy of the liquid at 1150 K.

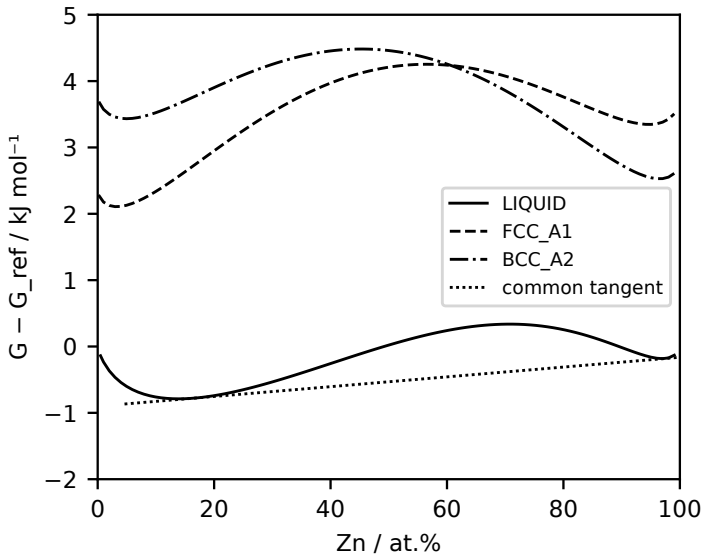


Fig. 4. Calculated Gibbs energy curves at 890 K.

Table 3. Redlich–Kister parameters of the liquid (Pb–Zn).

ν	a / J mol⁻¹	b / J mol⁻¹ K⁻¹	c / J mol⁻¹ K⁻¹
L0	32263.4	−13.10176	0.0
L1	−14909.2	10.92958	0.0
L2	2829.4	0.0	0.0

Table 4. Integral and partial mixing properties of the liquid at 1150 K (Pb–Zn).

at.% Zn	ΔmixH / kJ mol⁻¹	ΔmixG / kJ mol⁻¹	a(Pb)	a(Zn)
10	1.99	−1.57	0.9201	0.4114
20	3.89	−2.09	0.8644	0.5990
30	5.62	−2.33	0.8184	0.7081
40	7.05	−2.39	0.7763	0.7821
50	8.07	−2.33	0.7373	0.8333
60	8.49	−2.17	0.7036	0.8662
70	8.12	−1.94	0.6750	0.8860
80	6.76	−1.65	0.6377	0.9024
90	4.14	−1.23	0.5220	0.9318

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] T. Jantzen, P.J. Spencer, Calphad, 22, 417-34(1998).

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.